

Maintenance sheet version 9 for ablation area (tripod, one level)

Station name	KPC_U
People present	APA, CLAN, PHO, OEW + Kiddi & Bjarni
Purpose	Maintenance / <u>installation</u> / take-down / quick visit / emergency
Date	28/5-2022

Notes: Mark which serial number of PTA you install

must placed on top of snow

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	Y		171515		Clean? <u>Y</u> / N
Ablation hose assembly	Y		<u>50195</u> / 50164		No air in hose and bladder half full? <u>Y</u> / N
Iridium modem IMEI number	Y		300534063810550		
Inclinometer/compass	Y		Attached to the radiometer? <u>Y/N</u> <i>inside radium.</i>		
Satellite antenna	Y				
Wind sensor	Y		Black box pointing toward the radiometer (i.e., along the boom)? <u>Y</u> / N <i>281 cm above snow</i>		
Temperature / humidity assembly	Y		Fan spins? Y / N Sounds, OK? Y / N <i>Lufft sensor</i>		
HygroClip T/hum assembly	N		NB: Rotronics (Y / <u>N</u>) could have a 40 °C offset. New or old type? NB: Lufft sensor (<u>Y</u> / N)		
Sonic ranger on AWS	Y		Old sensor with new membrane? Y / N <i>235 cm to snow</i> <u>42+49</u> <i>(to snow + depth)</i>		
Sonic ranger on stake(s)	Y		Old sensor with new membrane? Y / N NB: make sure not being buried next winter <u>42+49</u> →		
Thermistor string	Y		<i>surface marking exactly at surface of ice</i>		
Solar panel	Y		Clean? <u>Y</u> / N NB: is not effective when partially covered		
GPS antenna	Y		At top of logger enclosure? <u>Y</u> / N → on top of logger ~2cm below top of casing		
Data logger	Y		CR1000 or <u>CR1000X</u> ?		
Multiplexer	N		Clicking during measurement cycle? Y / N <i>NO MULTIP.</i>		
Logger enclosure	Y		Replaced (including everything inside)? <u>Y</u> / N New desiccant bags? <u>Y</u> / <u>N</u> <i>NO DESSICANT BAGS</i>		
Battery box	Y		New box? <u>Y</u> / N New desiccant bags? <u>Y</u> / <u>N</u> <i>NO DESS. BAGS</i>		
Stakes	Y		New one(s) drilled in? <u>Y</u> / N Length of new stake(s) 8 m? <u>Y</u> / N Bottom plugged to reduce melt-in? <u>Y</u> / <u>N</u> <i>carbon smart stake</i>		
Tripod	Y		New tripod? <u>Y</u> / N New sensor boom? <u>Y</u> / N		

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)		0
Download data and/or change CF/SD Card (wait for green light)	Y / N	Y / N
Name of logger program		
Screen dump of values in fast scan mode	Y / N	Y / N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N	Y / N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	°
Mast tilt across boom looking from radiometer (- if clockwise)	°	°
Boom direction relative to north	° true/magn.	° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y / N	Y / N
Radiometer exactly aligned with inclinometer? If not, measure.	Y / N	Y / N
Temperature / humidity height (bottom of casing – surface)	cm	cm
Stake assembly standing?	Y / N / sort of	Y / N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	cm	42 + 50 cm
Free length of stake 2 (if 3: middle)	cm	— cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	— cm
Height of ablation bladder fluid level above the ice surface	cm	158 + 55 cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	snow covered cm
Length of thermistor string on surface from surface marking	cm	0 cm
Apparent damage to sensors? If yes, which?	Y / N	Y / N

Notes:

Ablation hose 13.6 m ^{depth} from ice surface
assuming markings are all in
Hose fluid ^{only} 106 cm above snow surface
Bladder empty

Must-do at next visit: